Milestone3 Progress Report

MedAssist AI: Medical Symptom Analysis & Disease Prediction System

Submitted By : Kanuri Hema Sri

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Project Title: MedAssistAI – Medical Symptom Analysis & Disease Prediction System

Intern Name: Kanuri Hema Sri

Mentor: Jyoteesh (Data Scientist)

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1.Introduction

The objective of Milestone 3 was to integrate the Machine Learning–based Risk Assessment module with the Disease Prediction module. The work focused on implementing ML model inference, preprocessing patient data using the BRFSS dataset, integrating the Disease Prediction API, generating final health risk assessments, and validating the complete workflow through testing. This aligns with the project's Milestone 3 goal of developing recommendation workflows and healthcare intelligence features.

2. Work Completed

2.1 ML Model Integration

Integrated the trained XGBoost Risk Assessment model into the FastAPI backend for patient health risk prediction.

2.2 Feature Preprocessing

Developed the preprocessing module to:

- Load the trained model artifact
- Load the imputer
- Convert patient information into BRFSS feature format
- Handle missing values before ML inference

2.3 Disease Prediction API Integration

Integrated the internal Disease Prediction API.

Implemented the Prediction Client to:

- Send patient symptoms
- Receive predicted disease and prediction confidence
- Handle API communication failures gracefully

2.4 Risk Assessment Workflow

Implemented the complete Risk Assessment workflow by:

- Receiving patient information
- Calling the Disease Prediction API
- Running the XGBoost model
- Generating ML risk probability

2.5 Decision Engine

Implemented the Decision Engine to:

- Combine ML prediction results with Disease Prediction outputs
- Calculate final risk score
- Determine risk level
- Determine severity
- Generate emergency alerts
- Generate healthcare recommendations

2.6 API Enhancement

Enhanced the

POST /risk-assessment

endpoint to return:

- Risk Probability
- Risk Score
- Risk Level
- Severity
- Emergency Alert
- Recommendations

while keeping Disease Prediction results internal.

2.7 Testing

Successfully tested:

- ML model inference
- Disease Prediction API integration
- Decision Engine
- Risk Assessment API
- End-to-end backend workflow

3. Technologies Used

Backend

- FastAPI
- Python

Machine Learning

- XGBoost
- Scikit-learn
- Joblib

Data Processing

- Pandas
- NumPy

Validation

- Pydantic

API Communication

- Requests

Testing

- Swagger UI
- Unit Testing

Version Control

- Git & GitHub

4. Current Module Status

Module	**Status**
ML Model Integration	Completed
BRFSS Feature Preprocessing	Completed
Disease Prediction API Integration	Completed
Prediction Client	Completed
Decision Engine	Completed
Risk Assessment API	Completed
API Testing	Completed

Integration Testing	Completed

5. Challenges Faced

The main challenges included integrating the Disease Prediction API with the ML-based Risk Assessment module, preprocessing BRFSS patient features correctly before inference, handling API communication failures, and combining multiple service outputs into a unified risk assessment. These challenges were addressed through a modular backend architecture, service-based implementation, and comprehensive testing.

6. Next Milestone

The next phase will focus on:

- Final system validation and performance optimization.
- Cloud deployment and Docker-based deployment.
- User Acceptance Testing (UAT).
- Final project demonstration and presentation.
- Project submission and deployment.

7. Conclusion

Milestone 3 successfully completed the integration of the MedAssistAI system by connecting the frontend, backend, database, Machine Learning Risk Assessment model, and Disease Prediction module into a unified workflow. The Risk Assessment module now processes patient information, performs ML-based risk prediction using the BRFSS dataset, integrates disease prediction results, and generates comprehensive health risk assessments with recommendations. The integrated system was validated through API, unit, and end-to-end testing, making it ready for the final deployment and project completion phase.